CYBERGUARD SENTINEL

AI-Based User Behavior Anomaly Detection Platform

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1. Executive Security Summary

This report summarizes behavioral telemetry, machine learning anomaly detection scores, and security rule evaluations across monitored user accounts. Anomalies are detected via Scikit-Learn Isolation Forest paired with behavioral baseline profiling and rule heuristics.

Total Monitored Events	ML Anomaly Detections	High-Risk Incidents	Open Security Alerts
41	5	5	6

2. Risk Score & Distribution Analysis

Risk Tier	Score Range	Event Count	Proportion (%)	Operational Policy
LOW RISK	0 - 29	35	85.4%	Standard logging; automatic baseline continuous learning
MEDIUM RISK	30 - 69	1	2.4%	Elevated monitoring; baseline freeze; SOC analyst triage queue
HIGH RISK	70 - 100	5	12.2%	Immediate alert raised; session termination / MFA step-up candidate

3. High-Risk & Suspicious Activity Log (Recent 8 Incidents)

Timestamp (UTC)	User	IP / Location	Target Resource	Risk Score	Flagged Detection Details
2026-10-31 20:06	admin	127.0.0.1 Internal Portal Ingress	/dashboard	**65.0** (Medium)	Login attempt at unusual hour (20:00) deviates from established user schedule.; Connection initiated via penetration testing OS or anonymized client: unknown os / unknown browser.
2026-10-31 20:06	alice.chen	45.154.255.89 External Suspicious Ingress	/api/v1/auth/login	**100.0** (High)	Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.77).; Login attempt at unusual hour (04:00) deviates from established user schedule.
2026-10-31 20:06	david.kim	103.152.220.10 External Suspicious Ingress	/financial/payroll	**100.0** (High)	Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.62).; Login attempt at unusual hour (23:00) deviates from established user schedule.
2026-10-31 20:06	eva.rossi	91.240.118.23 External Suspicious Ingress	/database/credentials	**100.0** (High)	Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.74).; Login attempt at unusual hour (01:00) deviates from established user schedule.

2026-10-31 20:06	bob.martinez	194.26.29.112 External Suspicious Ingress	/admin/system-logs	**100.0** (High)	Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.71).; Login attempt at unusual hour (03:00) deviates from established user schedule.
2026-10-31 20:06	alice.chen	185.220.101.5 External Suspicious Ingress	/api/v1/auth/login	**100.0** (High)	Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.79).; Login attempt at unusual hour (02:00) deviates from established user schedule.

4. MITRE ATT&CK; Adversarial Tactics & Technique Alignment

Security events, behavioral deviations, and Isolation Forest ML anomalies are correlated directly with official MITRE ATT&CK; Enterprise Matrix v14 tactics and techniques to facilitate standardized SOC response:

Technique ID	Tactic Category	Adversary Technique Name	Detection Heuristic & Indicator
T1110.001	Credential Access	Brute Force: Password Guessing	Adversary repeatedly attempts authentication passwords against valid account names to gain unauthorized access.
T1078.004	Defense Evasion & Initial Access	Valid Accounts: Cloud / Enterprise Accounts	Adversary leverages stolen credentials from geographically disparate locations, violating Haversine physical velocity limits.
T1083	Discovery & Credential Access	File and Directory Discovery	Adversary enumerates and probes sensitive application assets including credential stores, system configuration, and payroll vaults.
T1588.002	Resource Development	Obtain Capabilities: Penetration Testing Tool	Adversary interacts with authentication endpoints via adversarial distributions (Kali Linux, Parrot OS, Tor Browser) or automated scraping utilities.
T1071.001	Command and Control	Application Layer Protocol: Web Protocols	Adversary establishes authentication session through previously unobserved foreign IP, proxy, or anonymous ASN infrastructure.
T1078	Defense Evasion	Valid Accounts: Schedule Anomaly	Authentication attempted during anomalous off-hours schedule (e.g. 02:00 AM) to circumvent active administrator vigilance.
T1204	Execution	User Execution: Rogue Client Device	Session initiated from an unrecognized hardware device signature or spoofed User-Agent environment.
T1078	Initial Access	Valid Accounts: Multi-Vector Outlier	Isolation Forest identified extreme multi-dimensional divergence across timing, velocity, frequency, and duration dimensions.

5. Recommended Security Actions & Remediation

1. Multi-Factor Authentication (MFA) Step-Up: Enforce mandatory FIDO2 / TOTP challenge for logins originating from new IP addresses or unfamiliar client operating systems.

2. Account Lockout & Throttling: Configure automated temporary lockout or exponential rate-limiting for accounts accumulating 3+ failed logins in 5 minutes.

3. Baseline Safeguard Enforcement: Keep behavioral baseline profiles locked for users undergoing internal investigation to prevent adversarial profile poisoning.

4. Geofencing & Off-Hours Ingress Policies: Implement conditional access policies restricting administrative asset access between 23:00 and 06:00 UTC without prior SOC clearance.

5. SOC Incident Response: Review all OPEN High and Critical security alerts within a 15-minute SLA. Resolve alerts with thorough forensic notes.